

Model Comparison Report

Automated comparative analysis of evaluation summaries, thresholds, and per-slice performance.

This report compares 2 model outputs generated from structured evaluation summaries. The strongest model by macro F1 is **Qwen Mc Random** (0.4286), while the strongest model by exact accuracy is **Qwen Mc Transcription** (0.4975).

The lowest invalid-output rate is achieved by **Qwen Mc Random** (0.00% invalid predictions).

Across the current inputs, invalid predictions are the main limiting factor for every model, so coverage and output parsability are as important as raw classification quality.

Metric winners

Accuracy: Qwen Mc Transcription (0.4975)
Precision Macro: Qwen Mc Random (0.4969)
Recall Macro: Qwen Mc Transcription (0.4999)
F1 Macro: Qwen Mc Random (0.4286)
Coverage Rate: Qwen Mc Random (1.0000)
Invalid Rate: Qwen Mc Random (0.0000)

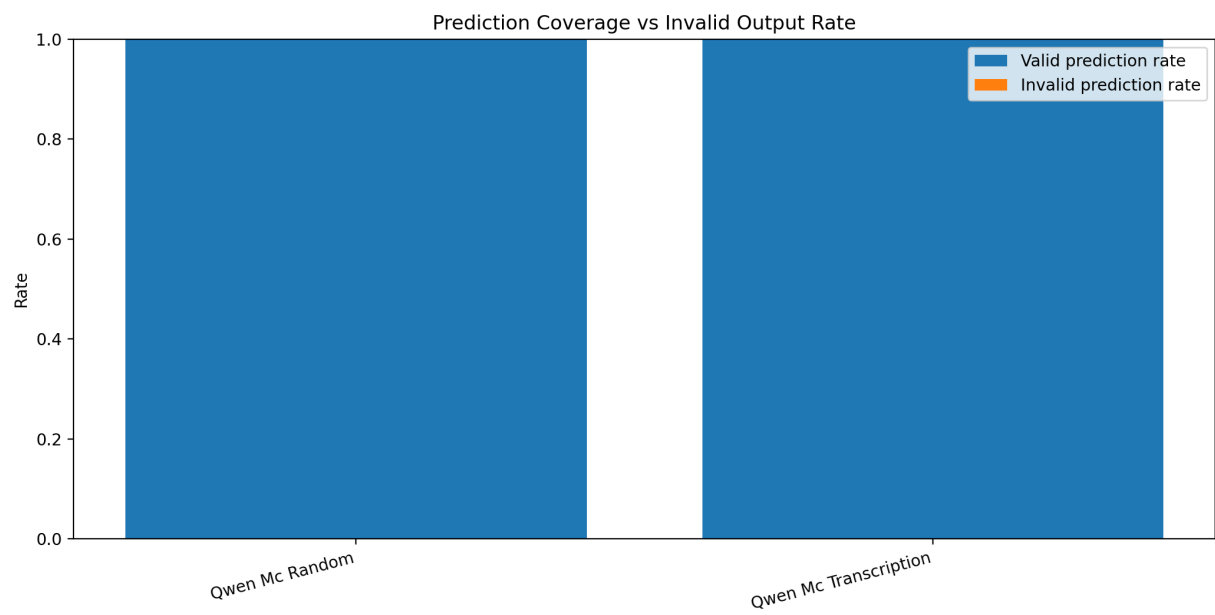
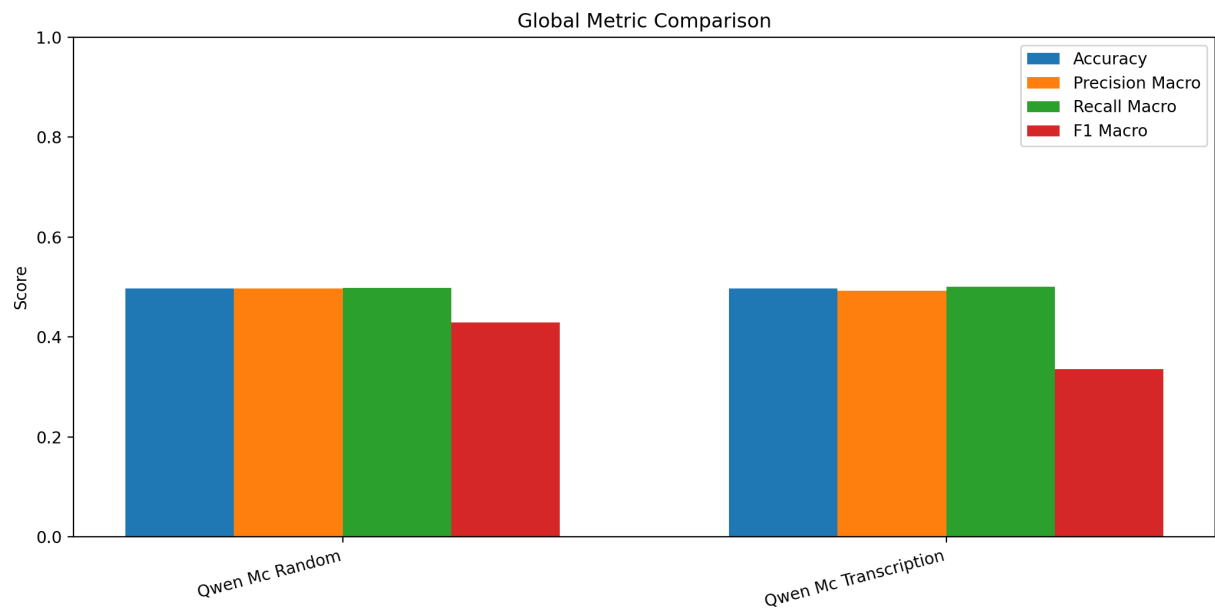
Input files

Model	Source File
Qwen Mc Transcription	qwen_mc_transcription_metrics_report.txt
Qwen Mc Random	qwen_mc_random_metrics_report.txt

Global comparison

The global table compares the most important aggregate indicators. Coverage rate is the proportion of parsable predictions, while invalid rate captures outputs that could not be mapped to labels 0 or 1.

Model	Accuracy	Precision Macro	Recall Macro	F1 Macro	Coverage Rate	Invalid Rate	Accuracy Band	F1 Band	Invalid Band
Qwen Mc Random	0.4967	0.4969	0.4984	0.4286	100.00%	0.00%	Moderate	Moderate	Excellent
Qwen Mc Transcription	0.4975	0.4922	0.4999	0.3356	100.00%	0.00%	Moderate	Moderate	Excellent



Threshold reference

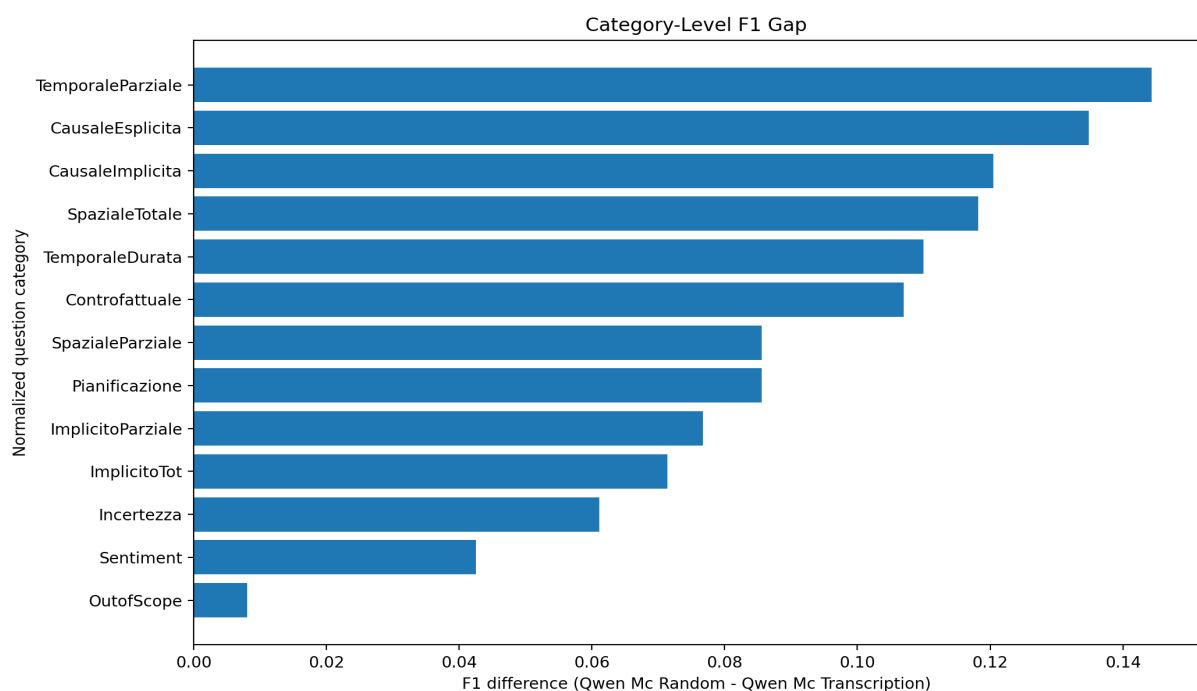
These thresholds are heuristic and intended for quick operational reading. They are not benchmark-specific scientific cut-offs, but they make the report easier to interpret consistently across runs.

Metric Family	Threshold	Interpretation
Accuracy / Precision / Recall / F1 / Coverage	≥ 0.75	Excellent
Accuracy / Precision / Recall / F1 / Coverage	0.50 - 0.7499	Strong
Accuracy / Precision / Recall / F1 / Coverage	0.25 - 0.4999	Moderate
Accuracy / Precision / Recall / F1 / Coverage	0.10 - 0.2499	Weak
Accuracy / Precision / Recall / F1 / Coverage	< 0.10	Critical
Invalid rate	≤ 0.05	Excellent
Invalid rate	0.0501 - 0.15	Acceptable
Invalid rate	0.1501 - 0.30	Concerning
Invalid rate	> 0.30	Critical

Category-level analysis

This section highlights which normalized question categories create the largest separation between models. Large F1 spreads indicate meaningful behavioral differences rather than minor noise.

Normalized Question Category	Best Model F1	F1 Spread
TemporaleParziale	Qwen Mc Random	0.1443
CausaleEsplicita	Qwen Mc Random	0.1349
CausaleImplicita	Qwen Mc Random	0.1205
SpazialeTotale	Qwen Mc Random	0.1182
TemporaleDurata	Qwen Mc Random	0.1100
Controfattuale	Qwen Mc Random	0.1070
Pianificazione	Qwen Mc Random	0.0856
SpazialeParziale	Qwen Mc Random	0.0856



Detailed category comparison. Wide tables are automatically split into compact blocks so every column remains fully visible.

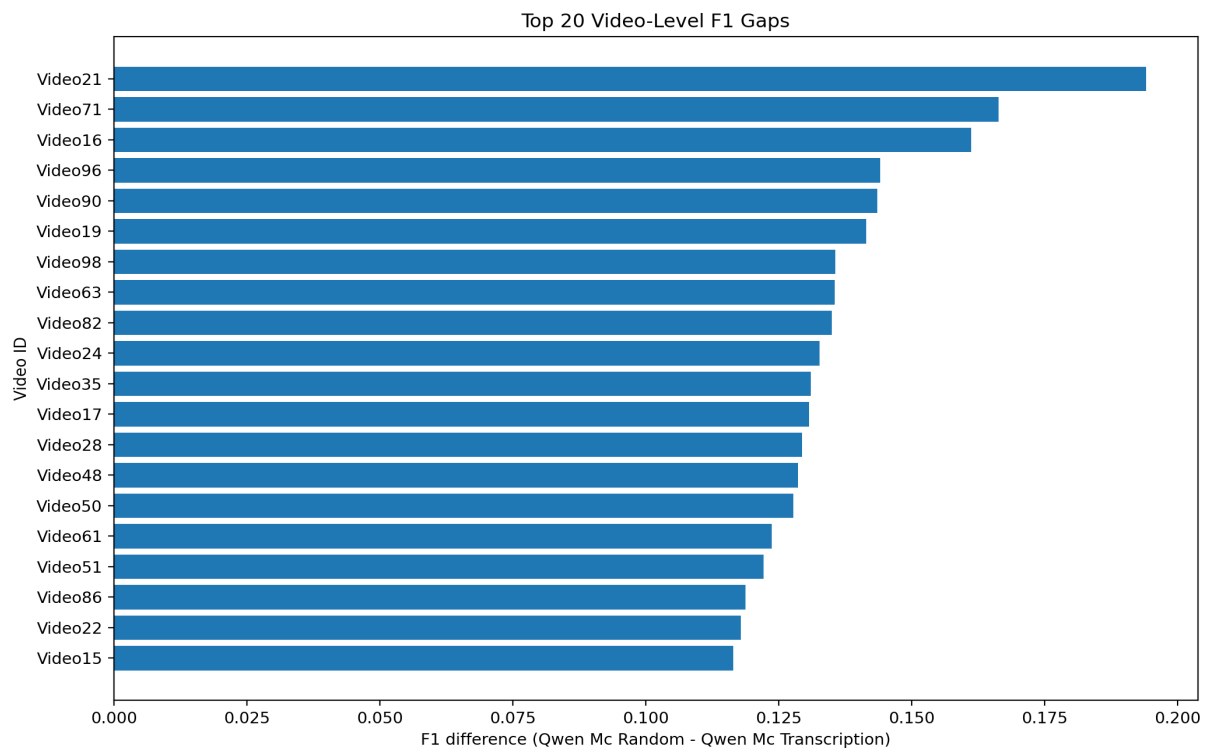
Category detail block 1

Normalized Question Category	Accuracy Qwen Mc Transcription	F1 Macro Qwen Mc Transcription	Invalid Rate Qwen Mc Transcription	Accuracy Qwen Mc Random	F1 Macro Qwen Mc Random	Invalid Rate Qwen Mc Random
TemporaleParziale	0.4800	0.3243	0.00%	0.4919	0.4686	0.00%
CausaleEsplicita	0.4850	0.3287	0.00%	0.4825	0.4636	0.00%
CausaleImplicita	0.5050	0.3355	0.00%	0.5025	0.4560	0.00%
SpazialeTotale	0.4925	0.3300	0.00%	0.5025	0.4482	0.00%
TemporaleDurata	0.5081	0.3369	0.00%	0.5062	0.4469	0.00%
Controfattuale	0.5125	0.3626	0.00%	0.4981	0.4696	0.00%
Pianificazione	0.5138	0.3394	0.00%	0.5094	0.4250	0.00%
SpazialeParziale	0.5075	0.3367	0.00%	0.4944	0.4223	0.00%
ImplicitoParziale	0.5025	0.3378	0.00%	0.5012	0.4145	0.00%
ImplicitoTot	0.5119	0.3453	0.00%	0.5100	0.4167	0.00%

Video-level analysis

Per-video comparisons help identify where performance differences are concentrated. A model can appear similar globally while behaving very differently on individual videos.

Video Id	F1 Spread
Video21	0.1941
Video71	0.1664
Video16	0.1612
Video96	0.1441
Video90	0.1436
Video19	0.1415
Video98	0.1356
Video63	0.1355
Video82	0.1350
Video24	0.1327
Video35	0.1311
Video17	0.1307



Detailed video comparison (top 15 rows by F1 spread).

Video detail block 1

Video Id	F1 Macro Qwen Mc Transcription	Invalid Rate Qwen Mc Transcription	F1 Macro Qwen Mc Random	Invalid Rate Qwen Mc Random
Video21	0.2941	0.00%	0.4882	0.00%
Video71	0.3447	0.00%	0.5111	0.00%
Video16	0.2967	0.00%	0.4579	0.00%
Video96	0.3287	0.00%	0.4728	0.00%
Video90	0.3661	0.00%	0.5097	0.00%

Video Id	F1 Macro Qwen Mc Transcription	Invalid Rate Qwen Mc Transcription	F1 Macro Qwen Mc Random	Invalid Rate Qwen Mc Random
Video19	0.3120	0.00%	0.4535	0.00%
Video98	0.3447	0.00%	0.4803	0.00%
Video63	0.3216	0.00%	0.4571	0.00%
Video82	0.3263	0.00%	0.4613	0.00%
Video24	0.3520	0.00%	0.4847	0.00%
Video35	0.3146	0.00%	0.4457	0.00%
Video17	0.3535	0.00%	0.4842	0.00%
Video28	0.3191	0.00%	0.4485	0.00%
Video48	0.3118	0.00%	0.4404	0.00%
Video50	0.3143	0.00%	0.4421	0.00%

Reading guide. When one model has higher accuracy but much lower coverage, the apparent gain may be misleading.

In these inputs, invalid prediction rate is the dominant operational bottleneck, so the most useful next step is to improve output formatting robustness first, then revisit class-level quality. Category-level spreads can guide targeted prompt tuning, decoding constraints, or post-processing rules.

Generated automatically from plain-text evaluation summaries using pandas, matplotlib, and ReportLab.